plain staining — dark and Light

عرفوا الكروموسومات عندما اكتشفوا المجهر

$1954 \to 46$ Chr

Elements of Medical Cytogenetics

all about chromosomes

Chapter 1

chromo → ملون
Som → جسم

chromosomes differ in all species

number of chromosomes does not reflect species size and shape

- Chromosome: chrom (colored) soma (body)
- 1954 recognition of 46 chromosomes not 48
- 1959 Down, Klinefelter, and Turner basis recognized

① klinefelter
② down Syndrome
③ Turner basis

Hypotonic: low concentration of salts

They put cells inside hypotonic solution so that leads to separation of chromosomes and then karyotyping.

karyotype → blood Lymphocytes "metaphase"

karyotyping

Banding
dark Light
identity

Banded chromosomes. Colored ones are also available!

Chromosomal morphology

A chromosome contains two arms (p and q) and a centromere → central part

After replication a chromosome is made of two sister chromatids

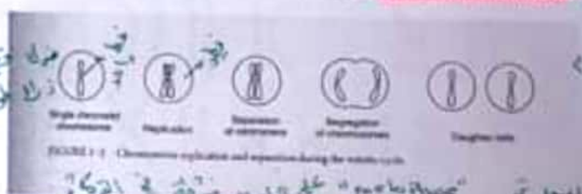


FIGURE 1-2 Chromosome replication and separation during the mitotic cycle.

"metaphase"

Routine classical cytogenetic analysis is done on mitotic representative blood lymphocytes "karyotype"

metaphase

Short → P [french]
long → q
arm

Centromere → Primary constriction

cytogenetic analysis

- Genome
- Homologous
- Autosomes (1-22)
- Sex chromosomes (Gonosomes)
- Haploid, gametocytes, ovum, sperm
- Heterozygote, homozygote

diff same

23 → sex chromosome
Gonosomes

Female → xx
male → xy

Chromosomes are distinguishable on the basis of their:

1. • Size → big or small
2. • Centromere position
3. • Banding pattern "barcode"

Centromere position

Centromere position → Metacentric, submetacentric, acrocentric, telocentric

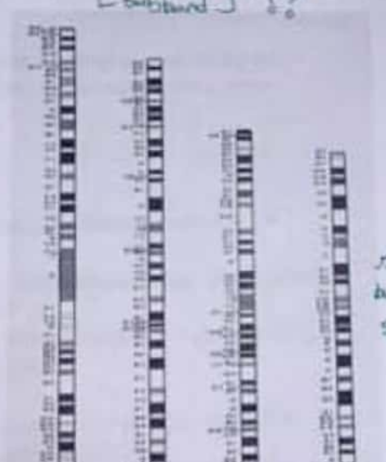
13, 14, 15 "Acrocentric"

1. Centromere
2. Size
3. banding pattern

only in another species

[band
subband] ??

Ideogram: diagrammatic representation of the banding pattern



* chr1 band3 region 1

Karyotype

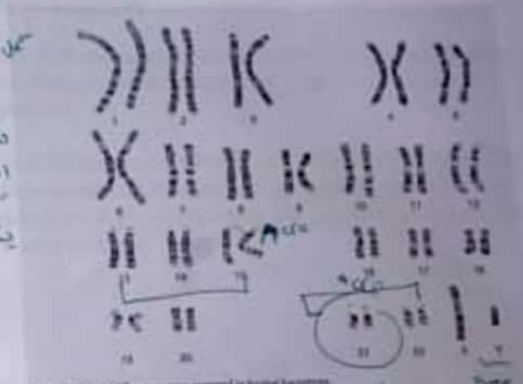


FIGURE 1-4 Chromosomes arranged in formal karyotype

Sex Chromosomes

"male"

1 Plain staining — dark and light

اقل وكو مفتوح وعليات جينات اكبر

Euchromatin: less condensed, contains coding DNA

Heterochromatin: more condensed, contains noncoding DNA

Telomeres: $\boxed{\text{TTAGGG}}$ repeated — نهاية الكروموسوم — يحفظ الكروموسوم

Centromeres multiple type — الجزء المركزي

Centromeric heterochromatin contains $\boxed{\text{satellite DNA}}$

From high expression to closed region — "position effect"

CHROMOSOME ABNORMALITY

① **Constitutional abnormality:** a chromosome abnormality that is present from conception and involves the entire body

② **Constitutional mosaicism:** if an additional cell line with a different chromosomal complement arises before the basis of the body structure is formed (i.e. in embryonic or pre-embryonic life) and becomes an integral part of the organism — embryo 8-9 weeks — 8-9

③ **Acquired chromosomal abnormality:** happens later on, after body structure is formed. — 8-9 weeks, 5-4 years

[Brain] got affected by genetic material dosage

disturbins → inhrited disease

An incorrect amount of genetic material carried out by the conceptus disturbs and distorts its normal growth pattern — conception

Trisomy ↑ 2+

Monosomy

Pathogenetic mechanisms that arise from chromosomal abnormalities: — Trisomy 21+

1. A dosage effect with a lack (deletion) or excess (duplication) of chromosomal material ①
2. A direct damaging effect, with disruption of a gene at the breakpoint of a rearrangement ②
3. Genomic imprinting ③
4. A position effect, whereby a gene in a new chromosomal environment functions inappropriately ③ — Level of expression

mono allelic expression ⇒

bi allelic expression → "abnormal" UPD

Poly ploidy

Full aneuploidy → presence of an abnormal number of chr

Partial aneuploidy

Loss of chromosomal material generally has a more devastating effect on growth of the conceptus than does an excess of material

Most full autosomal trisomies and virtually all full autosomal monosomies set development so awry that, sooner or later, abortion occurs – the embryo "self-destructs" and is expelled from the uterus

$\boxed{13, 14}$ TTAGGG

biallelic expression causes UPD

سموها ابزاء ملونة عشان على طول لما نطوا الصبغة اكتشفت الصبغة

Phenotype – first approach

To identify the chromosomal basis of many syndromes following analysis of groups of patients with similar phenotypes

Genotype – first approach

microarray, genome sequencing

New syndromes are being identified based on their DNA aberration primarily due to use of microarray analysis

13

Sex chromosomal abnormality

X inactivation: complementation

اذا صار لها activation راح يصير عندي overdose expression

Sex chromosomal imbalance has a much less deleterious effect on the phenotype than does autosomal aneuploidy

In both male and female, one, and only one, completely functioning X chromosome is needed

If an abnormal X chromosome is present, then, as a rule, cells containing this abnormal chromosome as the active X are selected against, perhaps due to preferential growth of those cells in which it is the normal X that is the active one

الخلايا اللي فيها x الخلل اله عشان يصير لها expression وتضاعف

14

Frequency and impact of cytogenetic pathology

Overall, around 1 in 135 liveborn babies have a chromosomal abnormality, and about 40% of these are phenotypically abnormal

ظاهرة دا منه عليها

If we were to look at 5-day blastocysts, the fraction with abnormality might be close to a half

If we studied a population of 70-year-olds, we could expect to see very few individuals with unbalanced autosomal karyotype

~~Karyotop~~ Karyotype: خلايا بتتضاعف at metaphase وتضيف عليها مواد كيماوية

x

Scanned by CamScanner

Chromosome Analysis

Chapter 2

① whole genome sequencing
② Classic analysis
③ Microarray

⑤ N.OR → Nucleus → Ribosomal RNA genes staining found in stacks

⑥ chromosomes in slides with Probe banded with fluorescent [Ab] detection for deletion, [redacted] replication

⑦ Looks like FISH, chromosomes on a slide "test DNA and Control DNA" Competition من غير Hybridization of

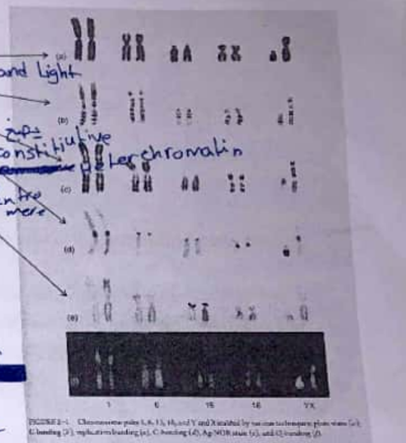
11:32 AM

1

تطلع من الكروموسوم مثل ما هو

Classic cytogenetic analysis

1. Plain staining
2. Giemsa banding — dark and Light
3. Replication banding — for barr body inactivation
4. Constitutive banding → بجيب constitutive heterochromatin in centromere
5. Nucleolar organizing regions staining
6. Fluorescence in situ hybridization
7. Comparative genomic hybridization



11:32 AM Chromosome rearragment

patient with Problems we use microarray analysis

2

Microarray analysis

تطلع على نقط

Chromosomal microarray is a first-line clinical diagnostic test for individuals with developmental disabilities

Classical cytogenetics is not likely to fade from view. Reasons:

1. Not all array results can give a definitive construction, and FISH is often necessary to elucidate the cytogenetics
2. The array cannot detect balanced rearrangements (though whole genome sequencing is being used to address this shortcoming)

Two types of microarrays:

1. Use a CGH approach
2. Use SNPs to detect the number of alleles in a specimen

11:32 AM

Isodisomy يعني كل الكروموسوم على monozygoes

3

SNP microarray

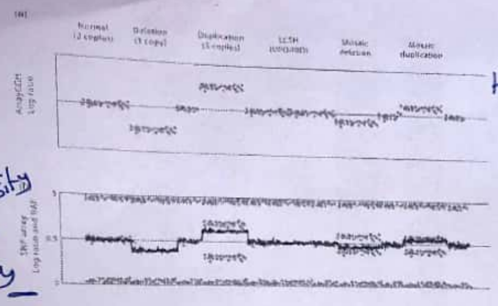
Heterozygous is distinguished from homozygous and from three alleles

- Homozygosity over a stretch of consecutive SNPs may indicate deletion — على جنبه يكون ال DNA
- Three alleles may indicate trisomy, ثلاث كروموسومات
- SNP microarray may detect uniparental disomy when the child's results are compared to the parental genotypes — كروموسومين من نفس الاب
- Isodisomy may be revealed, in the absence of parental samples, when the entire chromosome shows homozygosity

whole genome sequencing → بحسب الاكثرية gives overlabing fragments

11:32 AM

4



Heterozygosity ما في

tion → one copy

Human material

FIGURE 2-2 Interpretation of array comparative genomic hybridization (ArrayCGH) compared with SNP array: (a) In ArrayCGH, the signal between the test and reference sample is converted to a log ratio (gray dots) which acts as a proxy for copy number. An increase in log ratio represents a gain in copy number. (b) SNP arrays generate log ratio (black line) by comparing the signal intensities between different SNPs analyzed and offer an additional metric, the B allele frequency (BAF) (gray dots). The BAF is the component of the total allele signal (A + B) explained by a single allele (A). A BAF of 0 represents the genotype A/A or A/–, a BAF of 0.5 represents the genotype A/B, and a BAF of 1 represents the genotype B/B or B/–. Deletions result in the loss of heterozygous (A/B) genotypes, whereas duplications result in a separation of heterozygous genotypes into AA/B and A/BB. The BAF also allows for the detection of copy number neutral abnormalities such as uniparental disomy and identity by descent (IBD) which appear as absence of heterozygous (A/B) genotypes without change in the log ratio.

Sequence تحلل لكل ال

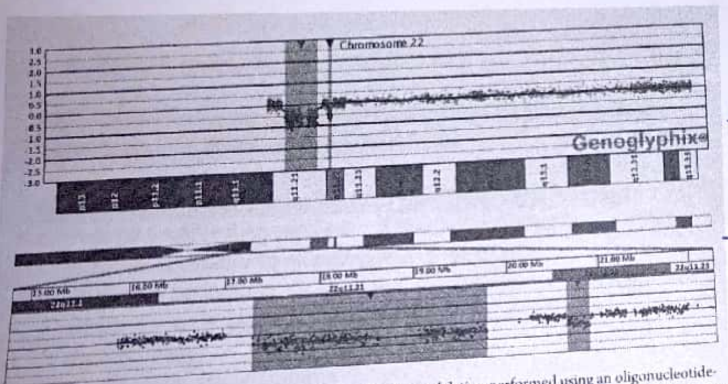
PCR-based applications

QF-PCR and **MLPA** are targeted approaches to assess DNA copy number

Next generation sequencing

Massively parallel genomic next generation sequencing

Expected a single NGS-based test will generate both copy-number and sequence data, and will become the first-line test for the investigation of children with developmental disabilities



microarray

FIGURE 2-3 Plot of chromosome 22 in a patient with a 22q11 deletion performed using an oligonucleotide-based array with comparative genomic hybridization. The deletion is indicated by the shaded area, which reflects a deviation from the $\log_2$ ratio of 1 (equal to zero). Distal to the classic diGeorge deletion is a common copy number variant (CNV).

Chromosomal findings from molecular analyses are often presented in an intuitive pictorial form

Higher resolution strategies may uncover DNA changes of unclear clinical significance, or unwanted information, such as loci that could predispose to cancer or adult-onset disorders

تغيرات changes in DNA

metaPhase

Origins and consequences of chromosome pathology

آليات انقسام الخلية

Chapter 3

Variation important for Species gives them value

males الانقسام مستمر في ال

11:33 AM

Meiosis in chromosomally normal persons

Diploid → Haploid & variation creation

Differences in timing between the two sexes

Synapsis crossing over

11:33 AM

- Synapsis
- Synaptonemal complex
- Bivalents
- Recombination
- Crossing-over

ينفصلوا عن بعض Desynapsis, dissociation of synaptonemal complex and chiasmata formation

مهم Stability for chr

Disjoin, disjunction, segregation

عدم الانفصال Nondisjunction

Malsegregation

11:33 AM

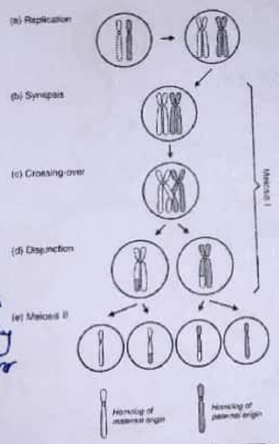


FIGURE 3-1 Chromosomal behavior during meiosis I, according to the classical model. [illegible]

Reverse segregation

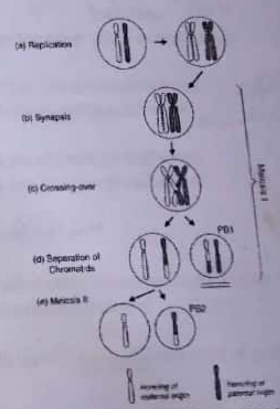


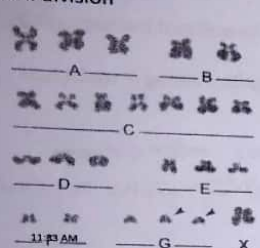
FIGURE 3-2 Chromosomal behavior during meiosis, [illegible] "reverse segregation." [illegible]

11:33 AM

Nondisjunction in meiosis

ترتيب خاطئ للكروموسومات

Nondisjunction: failure of homologous chromosomes or sister chromatids to segregate symmetrically at cell division



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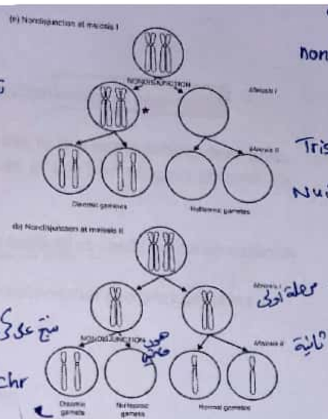


ضيع على كروموسوم واحد

2 chr

FIGURE 3-3 The classical view of the mechanics of nondisjunction. The asterisked gamete reflects the complement of the oöcyte in Fig. 3-4 (upper). In oogenesis, one of the two cells following meiosis I would be the first polar body, which might or might not proceed to meiosis II.

disjunction → انفصال

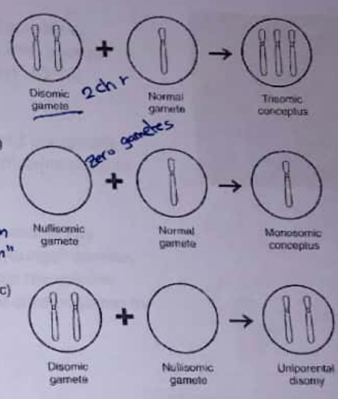
nondisjunction → ما راح ينفصل

Trisomy ⇒ 3

Nuisomy ⇒ ما في كروموسوم

مرحلة أولى → Homologous chromosome انفصال "reduction division"

ثانية → chromatids seperate



2 chr

zero gametes

11:33 AM FIGURE 3-5 Aneuploid gametes producing an aneuploid conceptus (a and b), and aneuploid gametes producing uniparental disomy (c).

oogensis gives Polar body

"Nondisjunction in meiosis I"

* Plain staining → cannot diff between chr's

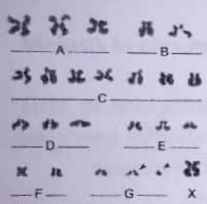


FIGURE 3-4 Oöcytes at metaphase of meiosis II, showing nondisjunction of a G-group chromosome having occurred at the preceding first meiotic division. Upper, oöcyte with classical nondisjunctional disomy, showing an additional G-group double chromatid chromosome. Possibly the arrowed pair are chromosome 21s, and the karyotype 24,X,+21. Lower, oöcyte with "predivisional" disomy, showing an additional G-group single chromatid. The arrowed pair may be chromosome 21s, and the karyotype 24,X,+21chr. (From Kamiguchi et al., 1993, courtesy ...)

Chromatid "predivision" hypothesis

receptors

risomy

11:33 AM

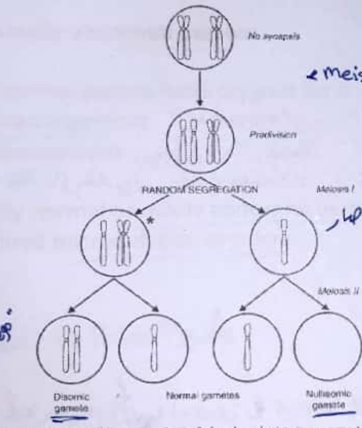


FIGURE 3-6 Nondisjunction following "predivision" of one homolog into its component chromatids in meiosis I (Angell's hypothesis). The asterisked gamete reflects the complement of the oöcyte in Figure 3-4 (lower). In oogenesis, one of the two cells following meiosis I would be the first polar body, which might or might not proceed to meiosis II.

meisois 2 ⇒ non sister chromatids or sister chromatids

* على حسب وين القطع صار

triploidy: ٣ مجموعات

tetra ploidy: 50

The majority of human malsegregation takes place in oogenesis

Women in their twenties, meiosis I errors are predominant

Women in their forties, meiosis II errors are predominant

UPD → الاثنين جايين من نفس الأب

Women aged 28-37, 47% of eggs are disomic or nullisomic

Women aged 38-47, 78% of eggs are disomic or nullisomic

لو ثلاث أرباع الـ egg

Multiple aneuploidy are known نفس الكروموسوم مجموعة

4 chr

Sequential nondisjunction at both meiotic divisions could lead to tetrasomy, this is the basis of some X chromosomal polysomy

69

Complete nondisjunction, could lead to triploidy, retention of polar body within the ovum

Simultaneous parental nondisjunctions a rarer source of double aneuploidy

11:33 AM *embryos have Large amount of abnormalities

Chromosomal abnormality in Zygote is very high

Causes of nondisjunction

Most aneuploidy due to nondisjunction $\boxed{\text{arises in oogenesis}}$

Vulnerability of maternal meiosis lies in the degradation, over time, of factors that support the adhesion of the homologous chromatids of the bivalent

Compromised function of spindle apparatus could cause aneuploidy

Mutations in genes coding for synaptonemal complex proteins

Synapsis → gives complex

فيه بروتينات كثير تساعد على الانفصال

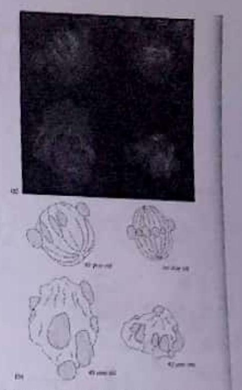
هالبروتينات مرات بصير فيها طفرات

11:33 AM

Researchers sampled oocytes at meiosis II metaphase from younger (20-25 years) and older (40-45) women

Looked at the nature of the spindle and the metaphase chromosomes as a whole

Found: symmetrical and neatly arrayed complex in younger women, while in older women the spindle was skewed and the chromosomes in a mess



11:33 AM

Tyrosome 2

Meiosis in chromosomally abnormal persons

First, phenotypically normal person heterozygous for a balanced structural rearrangement "inversion"

heterosynapsis, homosynapsis اعادة ترتيب x تقطع من ولا تنتقل من مكان تاني التغير إلى تحليل العقبي

Second, phenotypically normal or mildly abnormal person with molecularly-defined microdeletion or micro duplication

normal Person ⇒ Heterozygote

Hetero Synapsis ✱ اختلافية في قطعة ما ① بغيره وربط بداله

② ما قدرت على السكون المتطابقة

Homosynapsis ملتصق للمتطابقة

"egrolins"

11:33 AM

سواء في Nondisjunction in mitosis and the generation of mosaicism

mitosis or miosies

Mosaic

مجموعة من الخلايا معًا اخطاء وخلايا صحية هالاخطاء بتخليها تموت

Difference between mitotic and meiotic errors:
Mitotic produce mosaic conceptus, meiotic produce a nonmosaic abnormality

The earlier in embryogenesis that a mitotic error occurs, the greater the likelihood for a substantial fraction of the soma to be aneuploid, leading to increasing departure from normality of the phenotype

كل ما حصل في مرحلة مبكرة يكون صح

تكون اكبر abnormality
كل ما زادت الخلايا الخربانة
وتعرضنا للخطر

11:33 AM

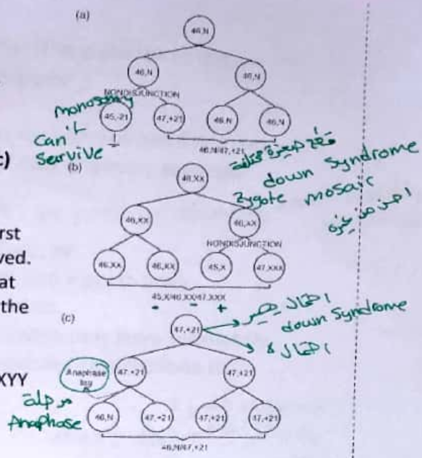
Nondisjunction in mitosis

Mitotic nondisjunction is the major mechanism causing mosaicism

(a) and **(b)** initially normal zygote, **(c)** initially aneuploid zygote

In **(b)**, if nondisjunction happen at first division then **no** 46,XX will be observed. Having some cells 46,XX indicates that nondisjunction happened later than the first division

Same applies to 46,XY with either X/XYY or X/XY/XYY



Generation of UPD as a result of trisomic rescue

Inference of rescue may be made in cases of UPD discovered because of isozygosity for a recessive gene

Postzygotic correction can happen in the other direction: to convert a monosomic zygote into a disomic one
45,XO to 46,XX

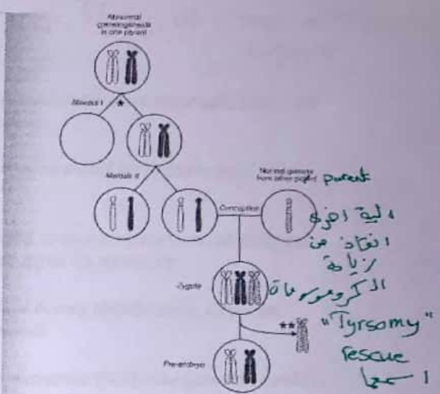


FIGURE 3-9 Uniparental disomy from "correction" of a trisomic conception by loss of a homolog. Nondisjunction at meiosis I, followed by postzygotic loss of one homolog, causes uniparental heterodisomy. (If for example, this were chromosome 15, and the meiotic nondisjunction occurred in the mother, the child would have Prader-Willi syndrome.) Nondisjunction at meiosis II would cause uniparental isodisomy.

monosomy → يصير اجهاض مباشرة
يصير لكن ما يكمل لمرحلة الولادة

Anaphase Lag [3 تصير خلية طبيعية
واحد من الكروموسومات تأخر في السحب وظل برا الـ Nuclear envelope

Myoder
راح يعطي
اشي اخف
complete down من
Syndrome

Non dis Junction happenes in meiosis 1 and 2
if happens in 1 ⇒ Heterodisomy
if happenes in 2 ⇒ uni Parental Isodisomy

Tysomy → موجود كل واحد من اب
وح جوزين من نفس الاب

" Autosomal recessive "
بس عند الاب والام ما عندها
فمن وين اجى الكروموسوم المصاب الاخر ؟

UPD
UPD heterodisomy
نفس الصورة
الكروموسومات مختلفة

UPD isodisomy
الكروموسومات متشابهة
Compentable isodisomy Tyrsomy

microdeletion كلواحد كان عنده

Epigenetic

motastic Pegus

DNA sequence of a particular gene remains unaltered, but the ability of this gene to be expressed is altered

Genomic imprinting " expression or no expression for allele "

Genes in the chromosomal segment are expressed, or not expressed, according to whether the chromosome had been transmitted in the sperm or in the ovum (parent-of-origin effect)

Biallelic gene expression is the norm

one allelic expression

Monoallelic gene expression is the exception (imprinted genes)

Genomic imprinting "only if one allele expressed"

The imprinting pattern may be specific to a certain tissue, or to a certain developmental stage

reversible

* Imprint is re-established in each subsequent generation depending of sex of parent

11:33 AM

21

Three categories of functional genetic defects connected to imprinting:

1. Uniparental disomy will lead to either biallelic expression, or no expression at the locus within the imprintable segment → expression from one parent only
2. Deletion at the imprinted locus
3. Relaxation of imprinting allows a segment that should be nonexpressed to lose its imprint

Biallelic expression

Trinucleotide repeat expansion

imprinting → كيف ممكن يسبب أمراض

UPD بتعمل مشاكل بس في ال imprinting

نقصان او زيادة بشكل كبير كل ما اثر سلبي وصار اجهاض بشكل مرحلة جدا مبكرة

11:33 AM

22

[Brain and Nerves]

Consequences of genetic abnormality

Structural imbalance

كل ما نقصت المادة قل التأثير السلبي

كل ما زادت زاد التأثير السلبي

Some genes are dosage sensitive

Incorrect amount of dosage-sensitive genetic material in every cell of the conceptus distorts its development to a greater or lesser extent

اذا أقل من ١٪ بعيش

اذا ازيد من ٢٪ بصير له اجهاض

Large losses or gains almost invariably set early development so awry that abortion occurs

داخل الرحم

Lesser imbalances may be compatible with continued intrauterine survival, but with the eventual production of phenotypically abnormal child

Mental defect is the almost universal consequence of autosomal imbalance

implance → جنين

11:33 AM

23

القدرة الكمية Quantitative assessment of structural imbalance

What is a large or a small degree of genetic imbalance

Quantitative – how much material is involved?

Haploid autosomal length (HAL) segments

تأثير على الـ development للانسان

Chromosome 1 comprises 8.5% of HAL

Chromosome 21 comprises 1.9% of HAL

If imbalance consists of less than 1% of HAL, the conceptus is often viable in utero, and live birth frequently results

If imbalance is greater than 2%, in utero lethality, with spontaneous abortion, is likely

Autosomal deficiency is much less survivable than duplication

Completive impertmance

11:33 AM

24

6

...ssment of structural imbalance

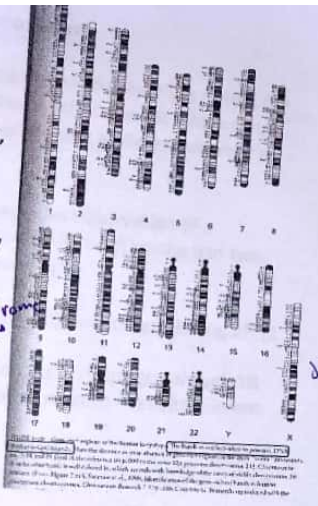
Some regions in the genome are more gene-rich than others, hence, abnormality herewith has more impact

توسيع الجينات على الكروموسوم
High genus

Trisomy 13, 3.7% of HAL, frequently goes through to live birth. 13q deletion syndrome 2.5% of HAL

syndrome
يعني

>> low gene density for chr 13



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Karyotype-phenotype correlations for deletions and duplications

Differing lengths of deleted or duplicated segments enable a dissection of the specific segmental contributions to components of an abnormal phenotype

A malformation map or deletion malformation map

Any X chromosomes in excess of one are genetically inactivated

45,X has a high in utero lethality

Phenocopies: similar phenotypes may flow from different genotypes
A non-genetic condition resembling a genetically determined one

11:33 AM

Phenocopy &

The mosaic state

milder → less severe

Whether mosaicism matters depends upon which tissue, and how much of that tissue, is abnormal

An individual with an aneuploid line in only some tissues is likely to have a less severe but qualitatively similar phenotype to some one with the nonmosaic aneuploidy

Some aneuploidies can only exist in the mosaic state, the nonmosaic form being lethal in utero e.g. trisomy 8

trisomy 8
Mosakismus

Mosaicism excluding the bone marrow will give a normal blood karyotype, and vice versa mosaicism confined to marrow would be seen on routine peripheral blood analysis, but not on other samplings

In sex chromosome mosaicism, fertility can exist when otherwise infertility is the rule e.g. Turner and Klinefelter

11:33 AM

HG

An interesting case of mosaicism for a structural rearrangement is that in which two lines of opposite imbalance coexist, with or without a normal cell line

Having no normal cell line suggest mosaicism resulted at the first mitosis of the zygote

Analysis of tissues other than blood can clarify the picture of mosaicism

infertil →

sterile → عقيم

11:33 AM

Autosomal reciprocal translocations

Chapter 5

denovo = Pure

translocation

Single double

Reciprocal translocation is an exchange of chromosomal material between non homologous chromosomes

Distal portion exchanged is the **translocated segment**
Rest of chromosome including centromere is the **centric segment**

Rearranged chromosome is called **derivative** (der) chromosome

1 person in 500 is a reciprocal translocation carrier (classic cyto)

Translocation may be de novo or run in families

chr5 chr5 der5 derelive chromosome

trans segment
centric segment

Microarray

قطعة صغيرة ① **Single segment exchange**: one of the translocated segments is
tip طرف جدا ~~very small~~ and comprises only the telomeric cap with the supposition of no genes contained
قطعة صغيرة ما عليها جينات

② **Double segment exchange**: both translocated segments are of substantial size
قطعة كبيرة فيها جينات متبادلة

ماتكل نادر **Whole-arm translocation**: rare and involves breakpoint at the centromere with an exchange of entire arms

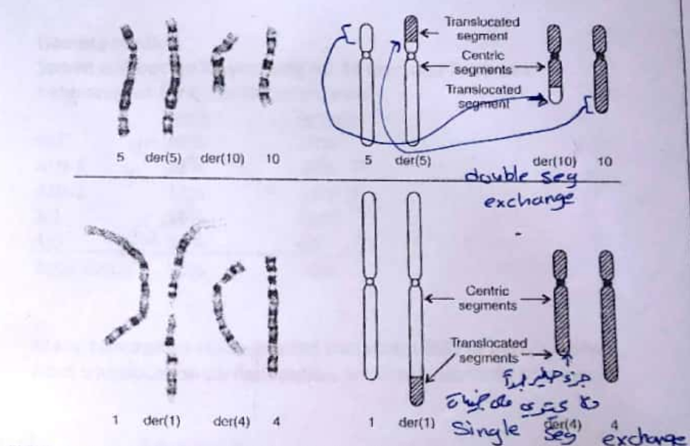


FIGURE 5–1 Reciprocal translocations demonstrating (*above*) double-segment and (*below*) single-segment exchange. The translocations are t(5;10)(p13;q23.3) and t(1;4)(q44;q31.3). (Cases of M. A. Leversha and N.

Scanned by CamScanner

لما تروح كلية
monosomy
trisomy
accours
خروج 2 وجود

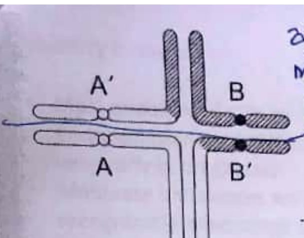


FIGURE 5–2 Pachytene configuration, simplified outline. The two normal (A, B) and the two translocation (A', B') homologs align corresponding segments of chromatin during meiosis I.

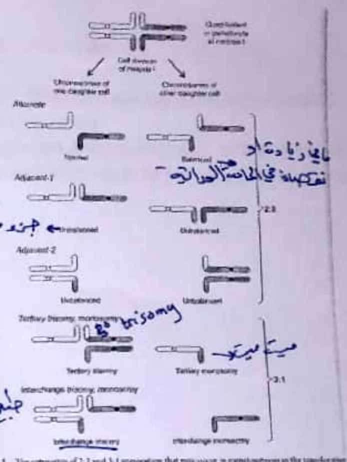
Quadrivalent

abnormal
gametes

Segregation

ONE DAUGHTER GAMETOCYTE WITH:	OTHER DAUGHTER GAMETOCYTE WITH:	SEGREGATION MODE
2:2 segregations		
A and B	A' and B'	Alternate segregation
A and B'	B and A'	Adjacent-1 segregation
A and A'	B and B'	Adjacent-2 segregation
3:1 segregations		
A B A'	B'	3:1 segregation with tertiary trisomy or monosomy
A B and B'	A'	
A' B' and A	B	3:1 segregation with interchange trisomy or monosomy
A' B' and B	A	
4:0 segregation		
A B A' B'	none	4:0 segregation with double trisomy or monosomy

جاميتا تكوينية وما فيها اشكال
الجنين بعيش
متبادل
بعيش
في كروموسوم
جنين



ما فيه زيادة او نقص في المادة الوراثية
جزء من كروموسوم زايد
جزء من كروموسوم ناقص
B trisomy
ميت ميت
جنين الزايد بدون خلقة

FIGURE 5–3 The categories of 2:2 and 3:1 segregation that may occur in gametogenesis in the translocation heterozygote. In the four 3:1 categories, only one of the two possible combinations in each category is depicted (both of each are shown in Fig. 5–4).

6

16 خيار
موجودين

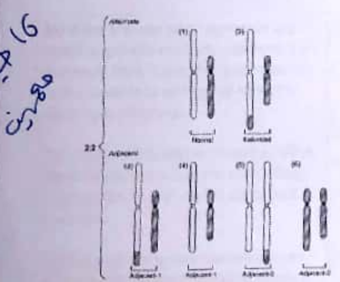
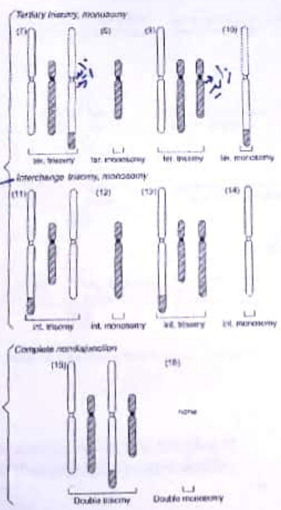


FIGURE 5–4 The full range of segregant gametes that can be produced by the translocation heterozygote, using the t(1;4) depicted in Figure 5-3 as an example. Chromosome 1 chromatin is shown open; chromosome 4 chromatin is crosshatched.

normal



بدون
Alternate
In a review of 1159 translocation families (1994), proportions of chromosomally unbalanced offspring were:
71% adjacent-1
4% adjacent-2
22% tertiary trisomy/monosomy
2.5% interchange trisomy

Gamete studies:
Sperm and oocyte karyotyping for 44 men and 7 women heterozygous for a translocation were:

	male	female
ALT	44 ~~43%~~	30%
ADJ-1	31 ~~34%~~	~~30%~~ 24
ADJ-2	13%	~~14%~~ 16
3:1	11 ~~10%~~	~~26%~~ 27
4:0	0.4 ~~~0%~~	~~0%~~ 2
Total abn.	55%	70%

Many **conception studies** found that about 80% of early embryos from translocation carrier couples are chromosomally abnormal

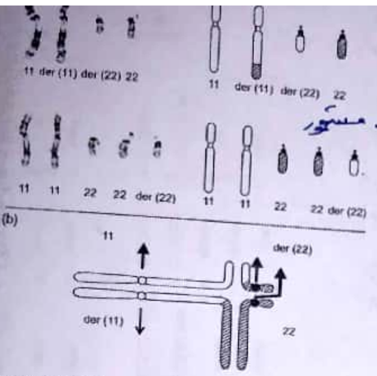


FIGURE 5-11 Tertiary trisomy. (a) The common t(11;22)(q23;q11) in the heterozygous state [illegible] the typical unbalanced state (below). (b) The presumed pachytene configuration during gametogenesis [illegible] heterozygote (chromosome 11 chromatin, open; chromosome 22 chromatin, crosshatched). Arrows [illegible] movements of chromosomes to daughter cells in a 3:1 tertiary segregation; heavy arrows show the [illegible] trisomic combination.

مرات يكون
خاص بعائلة معينة

3:1 Segregation with tertiary trisomy

t(11;22) is the spectacular exception to the rule that, in different families, translocations arise at different sites مشهور

The great majority of families have a "private translocation", and many may represent the first and only case in the whole of human evolution.

* تكرارها يدل على proximity

May reflect a physical proximity between the two chromosomes during meiosis في كيف زي

كروموسوم زايد في الحالة يومرات بعمل اعراض
مرات ما بعمل اعراض

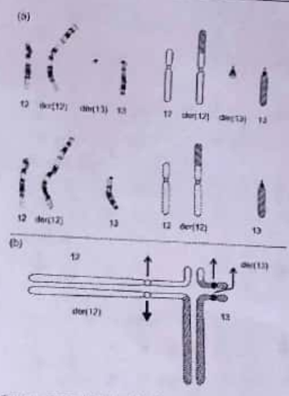


FIGURE 5-12 Tertiary monosomy. (a) Mother (above) has a reciprocal translocation between nos. 12 and 13, 46,t(12;13)(p13.3;q12.13). Two children (below) inherited the derivative 12, but no normal chromosome 12 or 13 from the mother, and have the karyotype 45,der(12). They are thus monosomic for the tip of 12p and pericentromeric 13 (and only a mildly abnormal phenotype). Chorionic villus sampling in a subsequent pregnancy gave a 46,XX result; an elder sister was a balanced carrier. (Case of M. D. [illegible].) (b) The presumed pachytene configuration during gametogenesis in the heterozygote (chromosome 12 chromatin, open; chromosome 13 chromatin, crosshatched). Arrows indicate movements of chromosomes to daughter cells in a 3:1 tertiary segregation; heavy arrow shows the monosomic complement. Alternatively, the three large chromosomes might form a trivalent, and the tiny der(13), being unattached, might segregate at random.

3:1 Segregation with tertiary monosomy

If one derivative is very small, and the amount of material that is missing is "monosomocally small", the countertype 3:1 22-chromosome gamete may lead to a viable conceptus

The large derivative chromosome is not far from being a composite of the two complete chromosomes

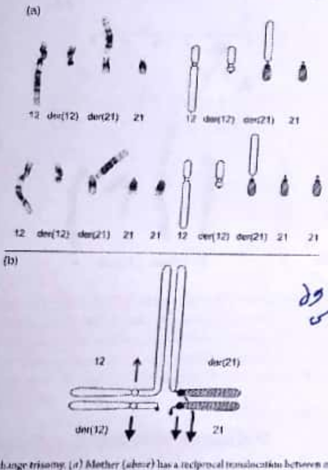


FIGURE 5-13 Interchange trisomy. (a) Mother (above) has a reciprocal translocation between nos. 12 and 21; her child (below) inherited the maternal translocation chromosomes and a "free" chromosome 21. The breakpoints are 12q13.1 and 21p13; an apparent gap, comprising satellite stalk, can be discerned between the centromere of the der(21) and its 12q component. (Case of R. Oertel.) (b) The presumed pachytene configuration during gametogenesis in the heterozygote (chromosome 12 chromatin, open; chromosome 21 chromatin, crosshatched). Arrows indicate movements of chromosomes to daughter cells in 3:1 interchange segregation, heavy arrows show the trisomic combination.

3:1 Segregation with interchange trisomy

This mode of segregation can only produce a live-born child when a "trisomically viable chromosome" (i.e. 13, 18, or 21) participate in the translocation

واحد من
هدول

الفرق بين tertiary and interchange

double trisomy : بس في

وضمن IVF

3:1 Segregation with interchange monosomy

Autosomal monosomy it typically associated with early arrested development

Only with PGD does a practical relevance possibly emerge

4:0 Segregation

Pre-implantation lethality would be the likely consequence

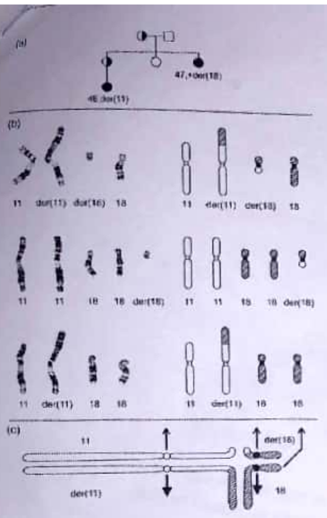
May be detected by PGD

translocation اذا حامل
يؤدي للجنين سليم

2:2 → alternate
→ Adjacent-1
→ Adjacent-2

3:1 → tertiary
→ inter change

4:0 في انواع



More than one unbalanced segregant type

This case is regarded a single-segment imbalance

- Adjacent-1 segregation
- 3:1 tertiary trisomy

(3:1 interchange trisomy is also possible)

More than one unbalanced segregant type

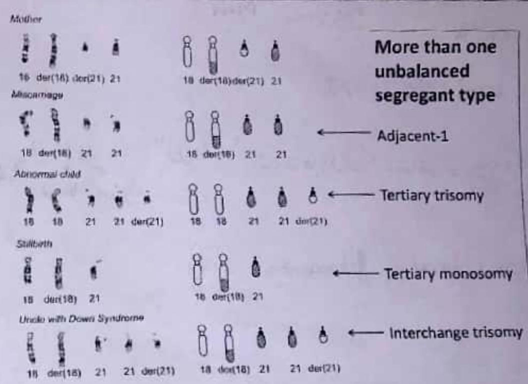


FIGURE 5–15 Several viable unbalanced forms. The karyotype is illustrated (*top*) of a mother carrying the translocation t(18;21)(q22.1;q11.2). She had a miscarriage due to adjacent-1 segregation, an abnormal child with a tertiary trisomy, and a stillborn child with a tertiary monosomy, as depicted in the cartoon karyotypes. An uncle with Down syndrome may have had the same adjacent-1 karyotype as in the second row, or possibly interchange trisomy 21, as depicted in the bottom row. (Case of M. D. Pertile.)

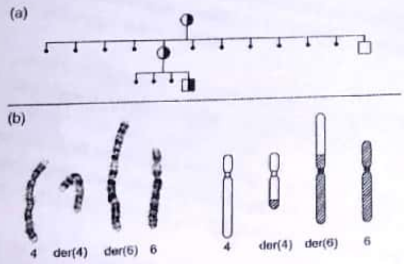


FIGURE 5–16 No unbalanced product viable. (*a*) Pedigree of a kindred in which mother and daughter have had multiple miscarriages, each having (*b*) the karyotype 46,XX,t(4;6)(q25;p23). (Case of A. J. Watt.) The presumed pachytene configuration during gametogenesis in the heterozygote would be as in Figure 5–5d (chromosome 4 chromatin, open; chromosome 6 chromatin, crosshatched) and, with large centric and translocated segments, the translocation has none of the features that enable viability of any unbalanced segregant combination.

No unbalanced mode possible

Long translocated and long centric segments
>> no mode of segregation could produce a viable unbalanced outcome

* different ways of segergation may lead to life born child